

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1510

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Code of Conduct

I ____ Shaik. Abdul Razak (192524358)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: __ Shaik. Abdul Razak _____

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

EXPERIMENT 1

Title

SIMATS Library Book Reservation System using Cloud SaaS

Aim

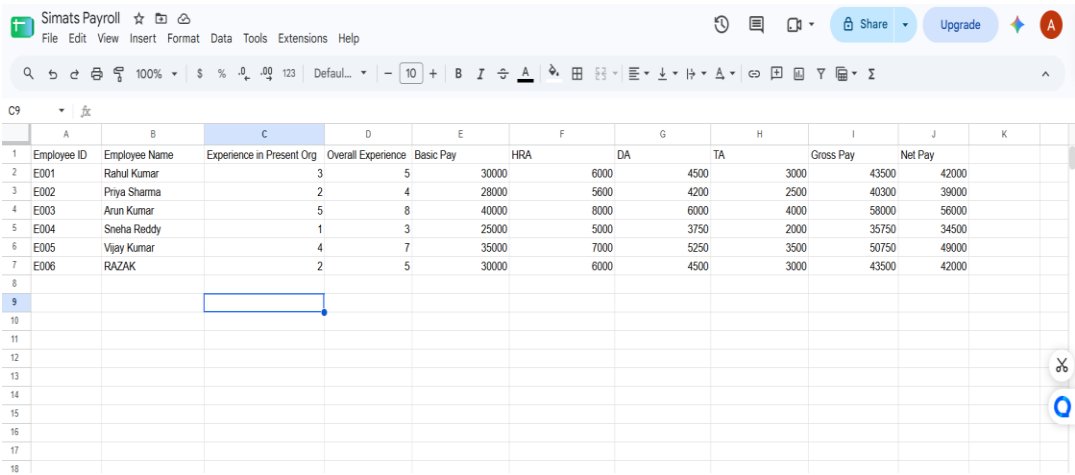
To create a simple cloud-based **Library Book Reservation System** using **Google Sheets and Google App Sheet** to demonstrate **Software as a Service (SaaS)**. The system manages book details such as Book ID, Title, Author, Publication, Year, Edition, Number of Copies, Rack Number, and Status.

Procedure

1. Create a Google Sheet named **SIMATS Library**.
2. Create columns for **Book ID, Title, Author, Publication, Year, Edition, No. of Copies, Rack No. and Status**.
3. Enter sample book records into the Google Sheet.
4. Open **Google App Sheet** and create an application using the Google Sheet as the data source.
5. Configure the columns and set **Status** as an **Enum** field.
6. Add two Status values: **Taken** and **Returned**.
7. Create a **Form View** for adding and reserving books.
8. Create a **Table/Deck View** to display the available library books.
9. Test the application by adding a book and changing its status.
10. Deploy the application and access it through a web browser or mobile device.

OUTPUTS:

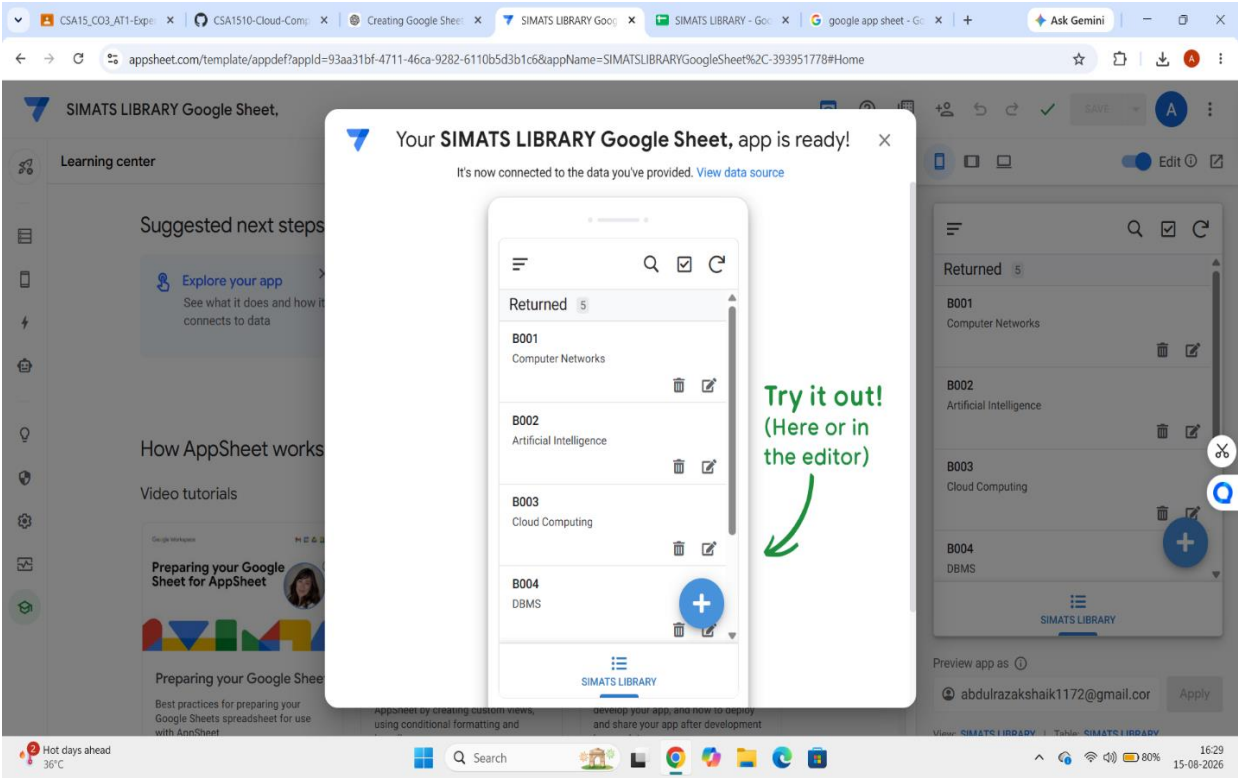
SIMATS Library Google Sheet with Sample Book Records:



The screenshot shows a Google Sheet titled "Simats Payroll" with a menu bar (File, Edit, View, Insert, Format, Data, Tools, Extensions, Help) and a toolbar. The spreadsheet contains a table with the following data:

Employee ID	Employee Name	Experience in Present Org	Overall Experience	Basic Pay	HRA	DA	TA	Gross Pay	Net Pay
E001	Rahul Kumar	3	5	30000	6000	4500	3000	43500	42000
E002	Priya Sharma	2	4	28000	5600	4200	2500	40300	39000
E003	Arun Kumar	5	8	40000	8000	6000	4000	58000	56000
E004	Sneha Reddy	1	3	25000	5000	3750	2000	35750	34500
E005	Vijay Kumar	4	7	35000	7000	5250	3500	50750	49000
E006	RAZAK	2	5	30000	6000	4500	3000	43500	42000

App Sheet Application Created from Google Sheet

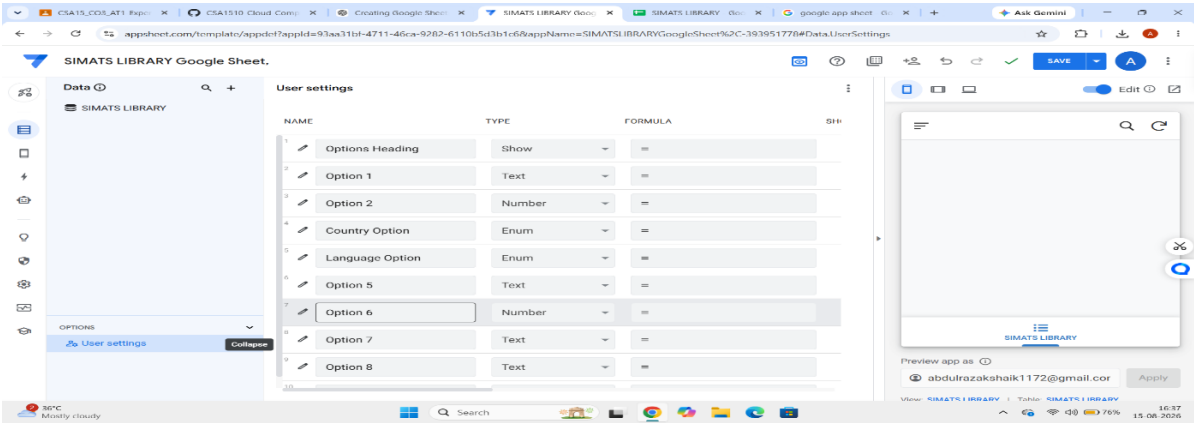


The screenshot shows the AppSheet application interface. A central white box displays a confirmation message: "Your SIMATS LIBRARY Google Sheet, app is ready! It's now connected to the data you've provided. View data source". Below the message is a preview of the app on a mobile device, showing a list of "Returned" items:

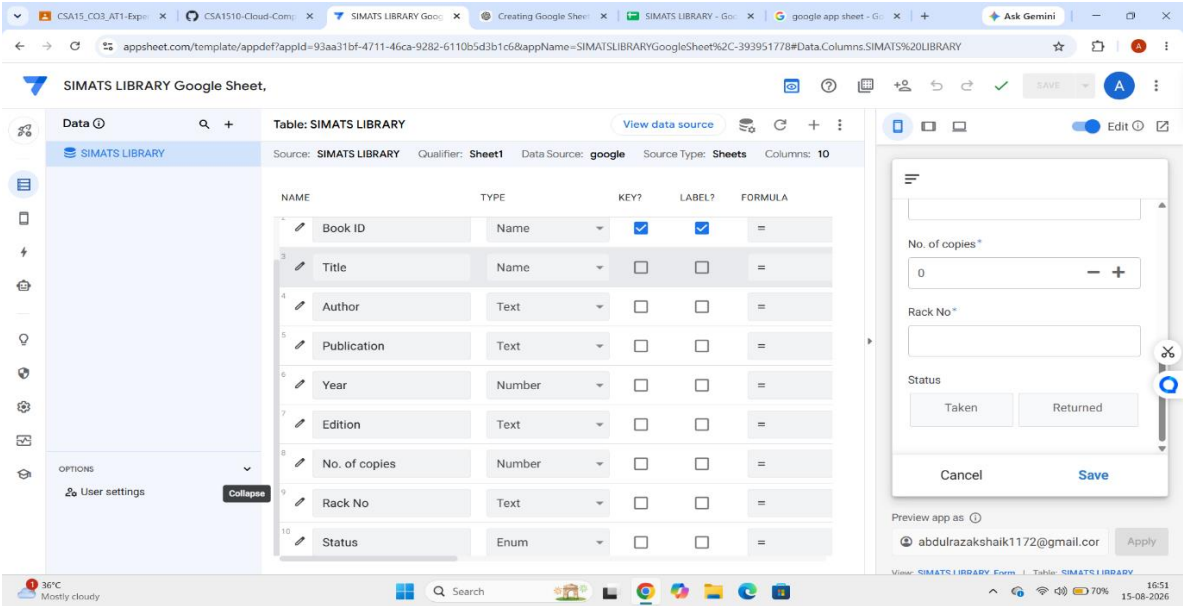
- B001 Computer Networks
- B002 Artificial Intelligence
- B003 Cloud Computing
- B004 DBMS

A green arrow points to the app preview with the text "Try it out! (Here or in the editor)". The background shows the AppSheet editor interface with a sidebar on the left and a list of items on the right. The bottom of the screen shows a Windows taskbar with the date and time: 16:29, 15-08-2026.

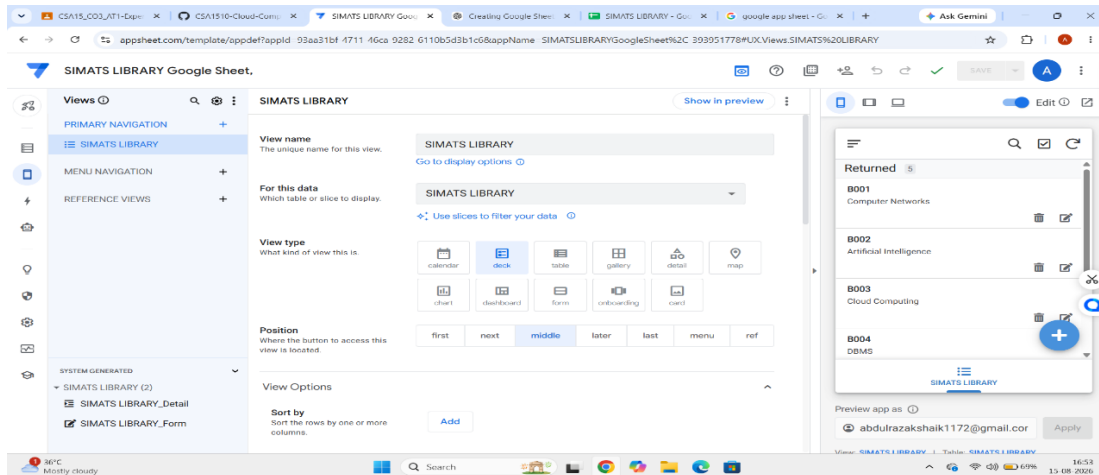
Database Column Configuration



Book Status Configuration — Taken / Returned



Library Application Preview and Book Records:



Result:

The cloud-based SIMATS Library Book Reservation System was successfully created using Google Sheets and Google AppSheet. Books were added and managed through the SaaS application, and the book status was successfully changed between Taken and Returned.

EXPERIMENT 2

Title

Payroll Processing System using Cloud SaaS

Aim

To create a simple cloud-based **Payroll Processing System** using **Google Sheets and Google AppSheet** to demonstrate **Software as a Service (SaaS)**. The system manages employee details such as experience, basic pay, HRA, DA, TA, gross pay, and net pay.

Procedure

1. Create a Google Sheet named **SIMATS Payroll**.
2. Create columns for **Employee ID, Employee Name, Experience in Present Org, Overall Experience, Basic Pay, HRA, DA, TA, Gross Pay, and Net Pay**.
3. Enter sample employee payroll records into the Google Sheet.
4. Open **Google AppSheet** and create an application using the Google Sheet as the data source.
5. Configure the payroll columns with appropriate data types.
6. Create a **Form View** for entering employee payroll details.
7. Create a **Table View** to display employee payroll records.
8. Enter sample employee details through the AppSheet form.
9. Save and verify the employee payroll record in the application.
10. View the complete payroll details of an employee.
11. Deploy the application and access it through a web browser or mobile device.

OUTPUTS:

SIMATS Payroll Google Sheet with Employee Records

Simats Payroll												
File Edit View Insert Format Data Tools Extensions Help												
100% 123 Default... 10 B I A												
	A	B	C	D	E	F	G	H	I	J	K	
1	Employee ID	Employee Name	Experience in Present Org	Overall Experience	Basic Pay	HRA	DA	TA	Gross Pay	Net Pay		
2	E001	Rahul Kumar	3	5	30000	6000	4500	3000	43500	42000		
3	E002	Priya Sharma	2	4	28000	5600	4200	2500	40300	39000		
4	E003	Arun Kumar	5	8	40000	8000	6000	4000	58000	56000		
5	E004	Sneha Reddy	1	3	25000	5000	3750	2000	35750	34500		
6	E005	Vijay Kumar	4	7	35000	7000	5250	3500	50750	49000		
7	E006	RAZAK	2	5	30000	6000	4500	3000	43500	42000		
8												
9												
10												
11												
12												
13												
14												
15												
16												
17												
18												

Payroll Database Column Configuration

Views

PRIMARY NAVIGATION

Simats Payroll

MENU NAVIGATION

REFERENCE VIEWS

SYSTEM GENERATED

Simats Payroll (2)

Simats Payroll_Detail

Simats Payroll_Form

Simats Payroll

View name

Simats Payroll

Go to display options

For this data

Simats Payroll

Use slices to filter your data

View type

calendar

deck

table

gallery

detail

map

chart

dashboard

form

onboarding

card

Position

first

next

middle

later

last

menu

ref

View Options

Sort by

Employee Name

Ascending

Employee ID*

3cefd661

Employee Name

Experience in Present Org*

0

Overall Experience*

Cancel

Save

Preview app as

abdulrazakshaik1172@gmail.cor

Apply

Payroll Employee List

Your **Simats Payroll** app is ready!

It's now connected to the data you've provided. [View data source](#)

Arun Kumar

E003

Priya Sharma

E002

Rahul Kumar

E001

Sneha Reddy

E004

Vijay Kumar

Try it out!

(Here or in the editor)

Complete Employee Details

Views

- PRIMARY NAVIGATION
 - Simats Payroll
- MENU NAVIGATION
- REFERENCE VIEWS

Simats Payroll

View name
The unique name for this view.
Simats Payroll
[Go to display options](#)

For this data
Which table or slice to display.
Simats Payroll
[Use slices to filter your data](#)

View type
What kind of view this is.
calendar, deck, table, gallery, detail, map, chart, dashboard, form, onboarding, card

Position
Where the button to access this view is located.
first, next, middle, later, last, menu, ref

View Options

Sort by
Sort the rows by one or more columns.
Employee Name, Ascending

Preview app as
abdulrazakshaik1172@gmail.cor

App Preview

Simats Payroll

View name
The unique name for this view.
Simats Payroll
[Go to display options](#)

For this data
Which table or slice to display.
Simats Payroll
[Use slices to filter your data](#)

View type
What kind of view this is.
calendar, deck, table, gallery, detail, map, chart, dashboard, form, onboarding, card

Position
Where the button to access this view is located.
first, next, middle, later, last, menu, ref

View Options

Sort by
Sort the rows by one or more columns.
Employee Name, Ascending

Preview app as
abdulrazakshaik1172@gmail.cor

Result:

The cloud-based Payroll Processing System was successfully created using Google Sheets and Google App Sheet. Employee payroll details including basic pay, HRA, DA, TA, gross pay, and net pay were successfully entered, stored, and displayed through the SaaS application.

EXPERIMENT 3: Demonstration of Virtualization Using Type-2 Hypervisor

AIM

To demonstrate virtualization by installing a Type-2 hypervisor, creating and configuring a Virtual Machine with a Host Operating System, and allocating 1 CPU, 2 GB RAM, and 15 GB storage.

SOFTWARE REQUIREMENTS

- Oracle VirtualBox
- Windows/Linux ISO image
- Computer/Laptop with virtualization support

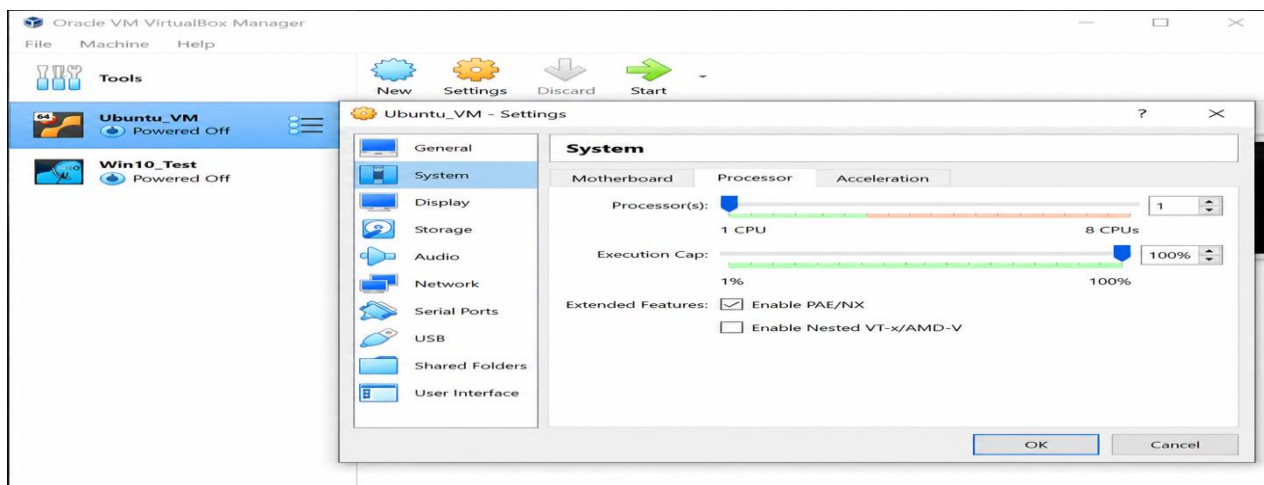
PROCEDURE

1. Download and install Oracle VirtualBox on the host computer.
2. Open VirtualBox and click New to create a Virtual Machine.
3. Enter a suitable name for the Virtual Machine.
4. Select the required Operating System and its version.
5. Allocate 2 GB (2048 MB) RAM to the Virtual Machine.
6. Configure the processor setting and allocate 1 CPU.
7. Create a new virtual hard disk with a size of 15 GB.
8. Select the required virtual disk type and storage option.
9. Attach the Windows/Linux ISO image to the Virtual Machine.
10. Start the Virtual Machine and install the Guest Operating System.
11. After installation, verify the Virtual Machine configuration.
12. Run the Virtual Machine successfully and observe the Guest Operating System.

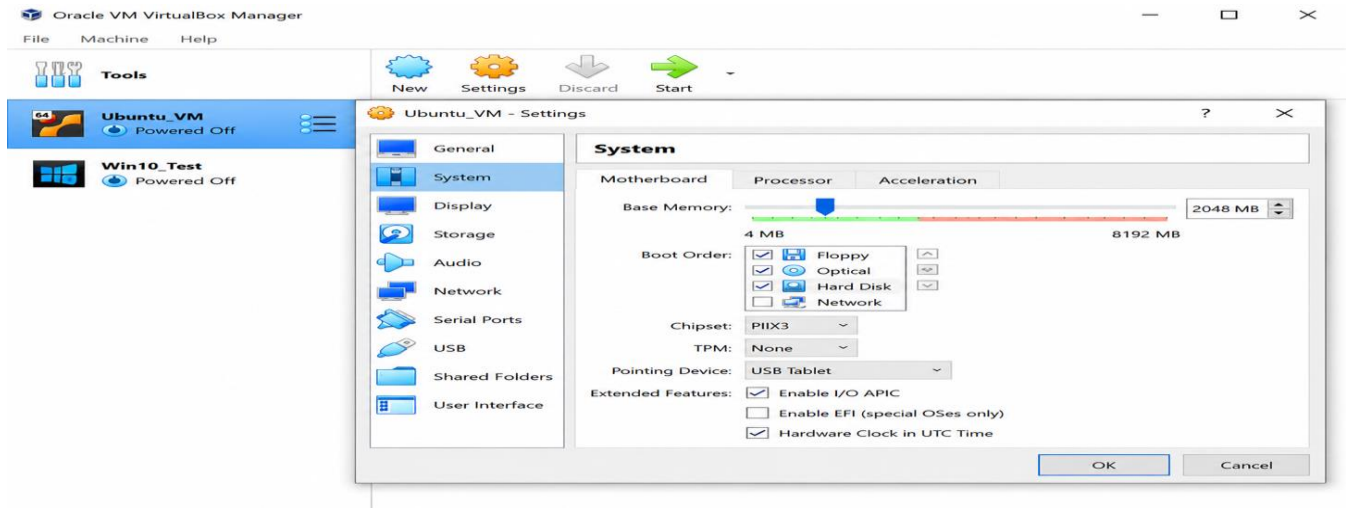
VM CONFIGURATION:

Parameter	Configuration
Hypervisor	Oracle VirtualBox
Virtualization Type	Type-2
CPU	1 CPU
RAM	2 GB
Storage	15 GB
Guest OS	Windows/Linux
Host OS	Windows/Linux

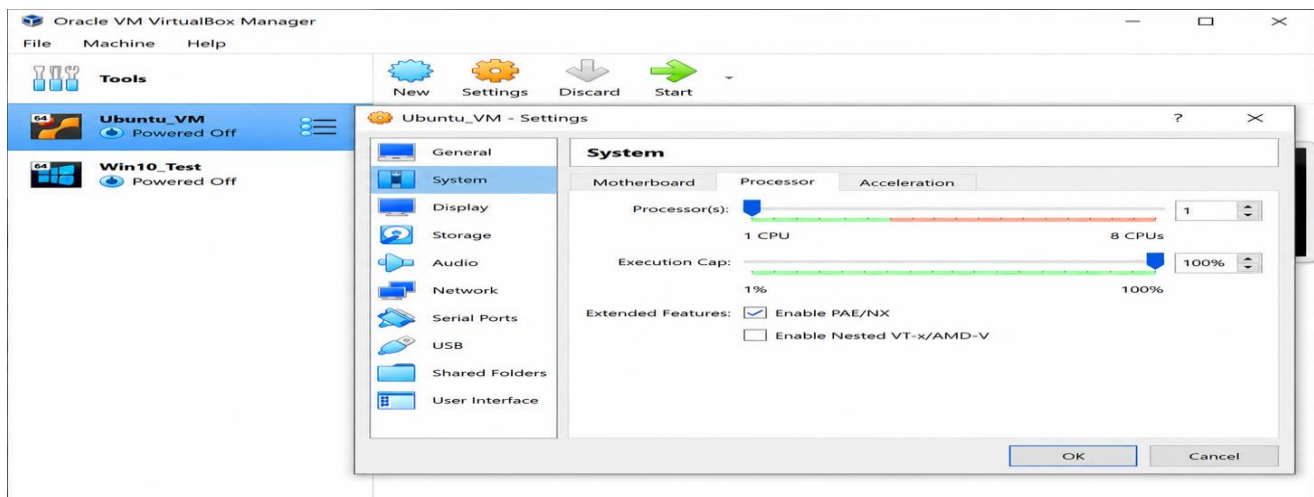
VM Creation / OS Selection



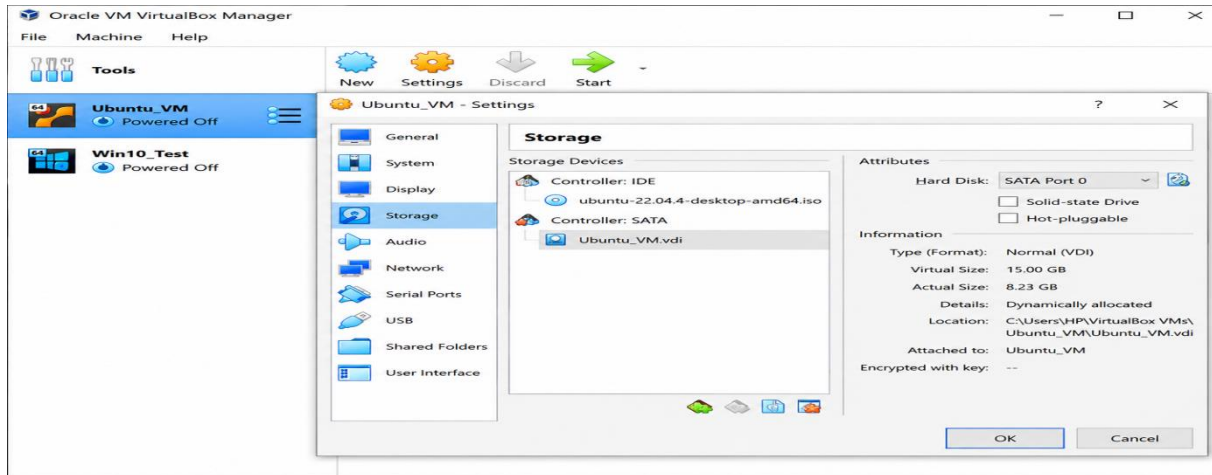
RAM Allocation – 2 GB



CPU Allocation – 1 CPU



Storage Allocation – 15 GB



RESULT

Thus, virtualization was successfully demonstrated using the Oracle VirtualBox Type-2 hypervisor. A Virtual Machine was created and configured with 1 CPU, 2 GB RAM, and 15 GB storage, and the Guest Operating System was successfully installed and executed.

EXPERIMENT 4: Cloning and Testing a Virtual Machine

AIM

To demonstrate virtualization using a Type-2 hypervisor by creating a clone of an existing Virtual Machine and testing the cloned VM by loading and running it.

SOFTWARE REQUIREMENTS

- Oracle VirtualBox
- Previously created Virtual Machine
- Windows/Linux Guest Operating System

PROCEDURE

1. Open Oracle VirtualBox.
2. Select the previously created Virtual Machine.
3. Make sure the original Virtual Machine is completely powered off.
4. Right-click the Virtual Machine and select Clone.
5. Enter a suitable name for the cloned Virtual Machine.
6. Select the cloning type as Full Clone.
7. Select the option to create new MAC addresses for network adapters if required.
8. Click Finish to create the cloned Virtual Machine.
9. Wait until the cloning process is completed.
10. The cloned Virtual Machine will appear separately in VirtualBox.
11. Start the cloned Virtual Machine.
12. Verify that the Guest Operating System loads successfully.
13. Test basic operations such as opening applications, accessing files, and checking the system configuration.

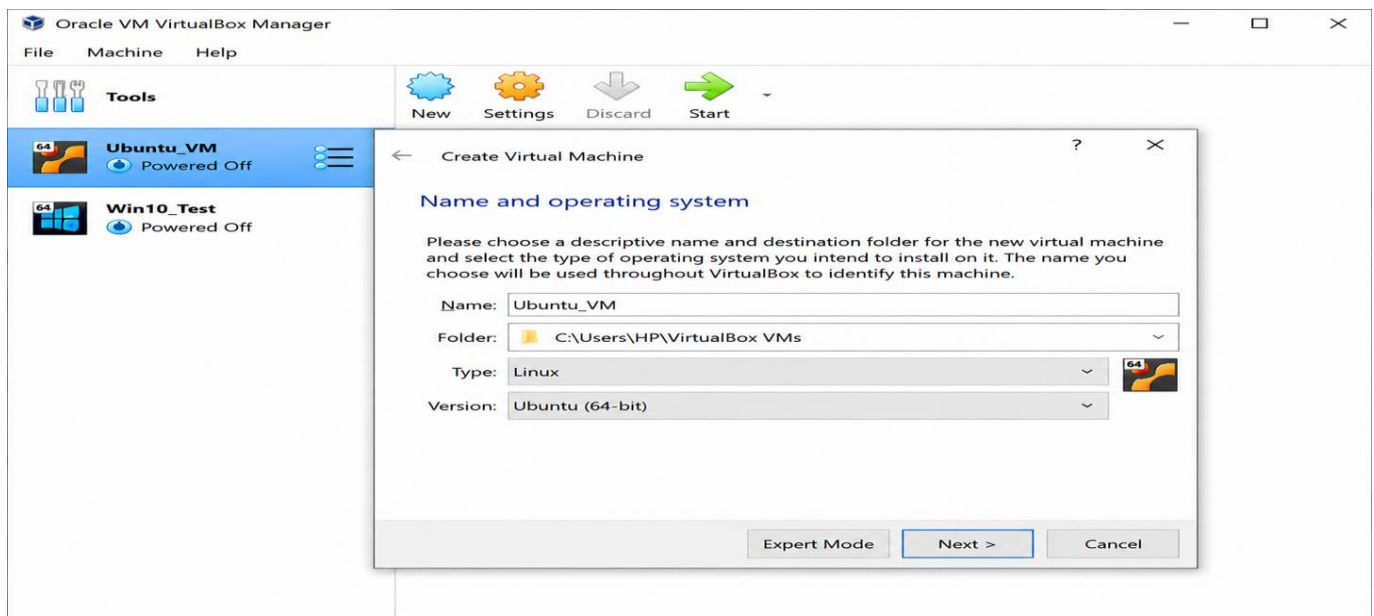
14. Shut down the cloned VM after successful testing.

15.Start the original VM again and verify that both the original and cloned VMs are available independently.

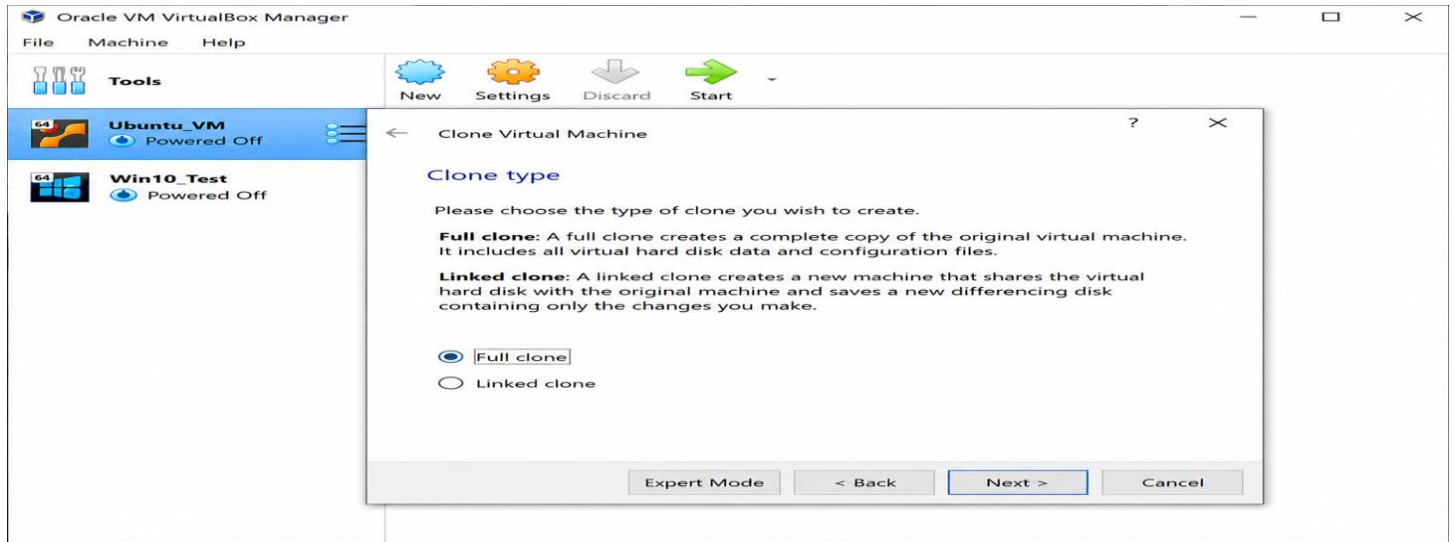
CLONING CONFIGURATION

Parameter	Configuration
Hypervisor	Oracle VirtualBox
Original VM	Previously Created VM
Clone Type	Full Clone
CPU	1 CPU
RAM	2 GB
Storage	15 GB
Guest OS	Windows/Linux

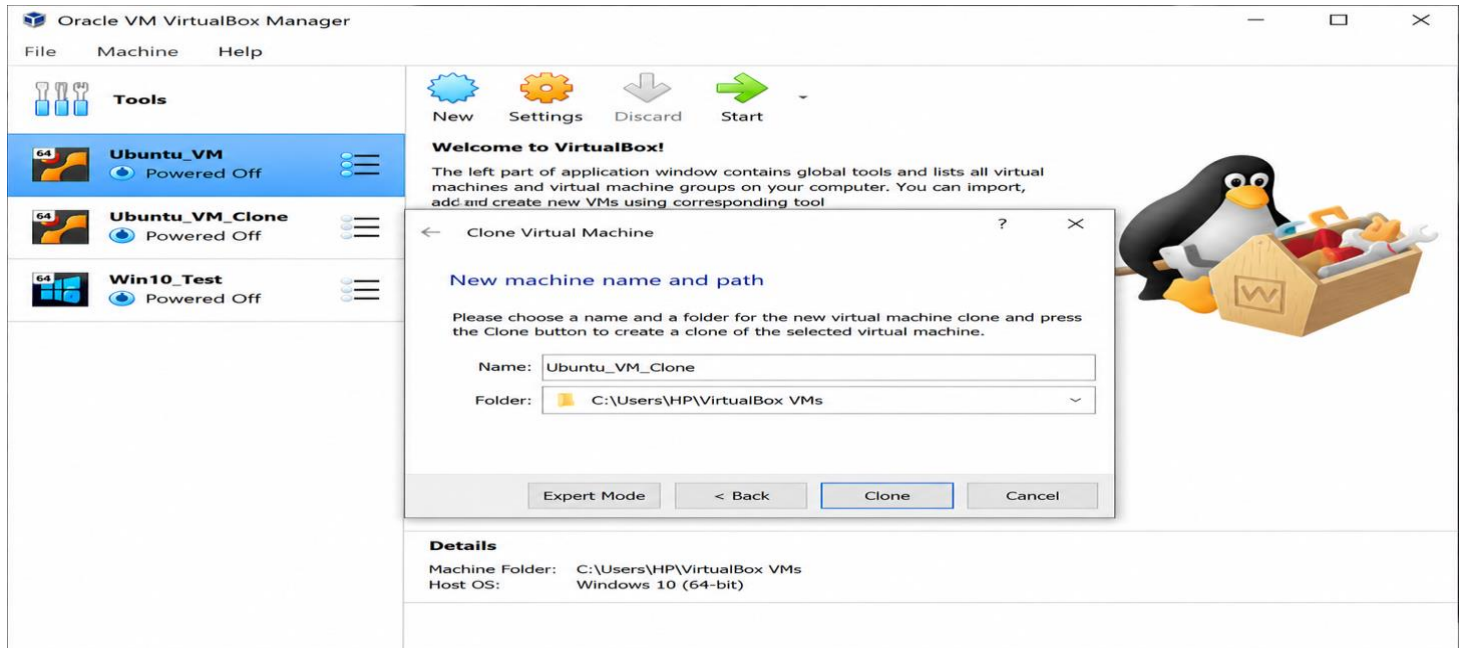
Selecting Clone Option from VirtualBox Manager



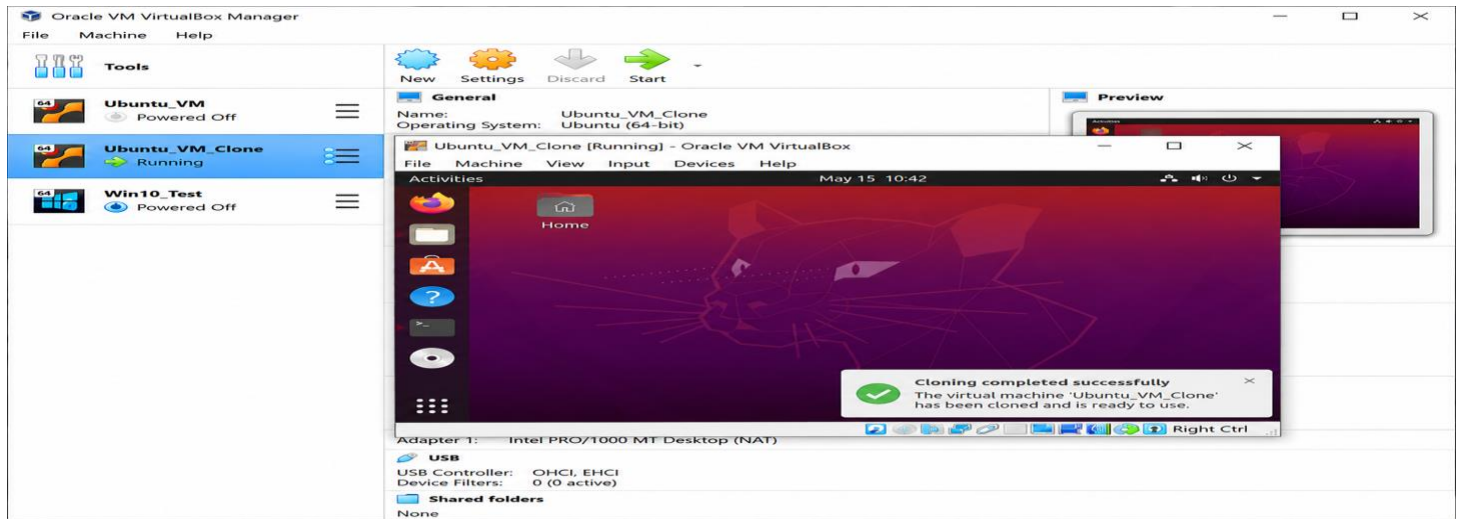
Selecting Full Clone Option



Entering Clone Name and Creating the Virtual Machine Clone



Cloned Virtual Machine Running Successfully



RESULT

Thus, a Virtual Machine was successfully cloned using the Oracle VirtualBox Type-2 hypervisor. The cloned VM was loaded and tested successfully, demonstrating that a VM can be duplicated and operated independently from the original VM.